Data Analyst Capstone Project

by Ainamkoz Aronkyzy

November 2026



© IBM Corporation. All rights reserved.

Skills Network

OUTLINE



- Executive Summary
- Introduction
- Methodology
- Results
  - Visualization – Charts
- Discussion
  - Findings & Implications
- Conclusion
- Appendix

EXECUTIVE SUMMARY



- The aim of this presentation is to summarize future skills requirements and trends in the IT sector to help the company to keep pace with changing technologies and make more informed hiring and developing decisions, remain competitive in the fast-evolving industry.

- This analysis is based on data from various sources including: job postings, blog posts and surveys. Collected data were prepared for analysis by using data wrangling methods, then statistical techniques were applied to identify insights and trends:
  - Top programming languages, database skills, platforms in demand
  - Trends for languages, databases, platforms for the next year
  - Demographic data like gender and age distribution of developers, education, level of compensation, their median values, distribution and comparison.

INTRODUCTION



- This presentation is a result of collecting and analyzing data from various sources, such as job postings on Naukri, IBM training's dataset, survey about popular languages and salaries in the industry. The most significant amount of data came from the Stack Overflow's annual Developer Survey 2019.

- The actual data set of the survey has around 90,000 responses. In this analysis a subset, that contains around 1/10th of the original data set (N = 11 398), was examined.

- This report is relevant for educators, students, developers, recruiters, HR specialists to understand the trends and skills requirement in the IT industry.

METHODOLOGY

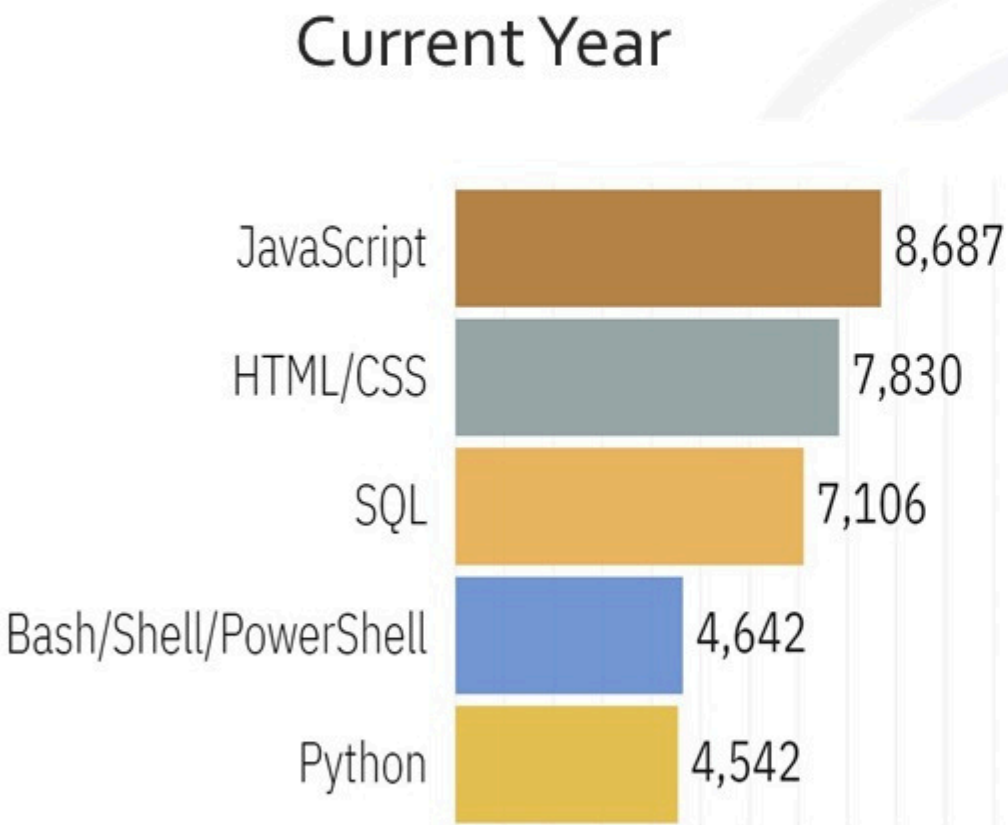


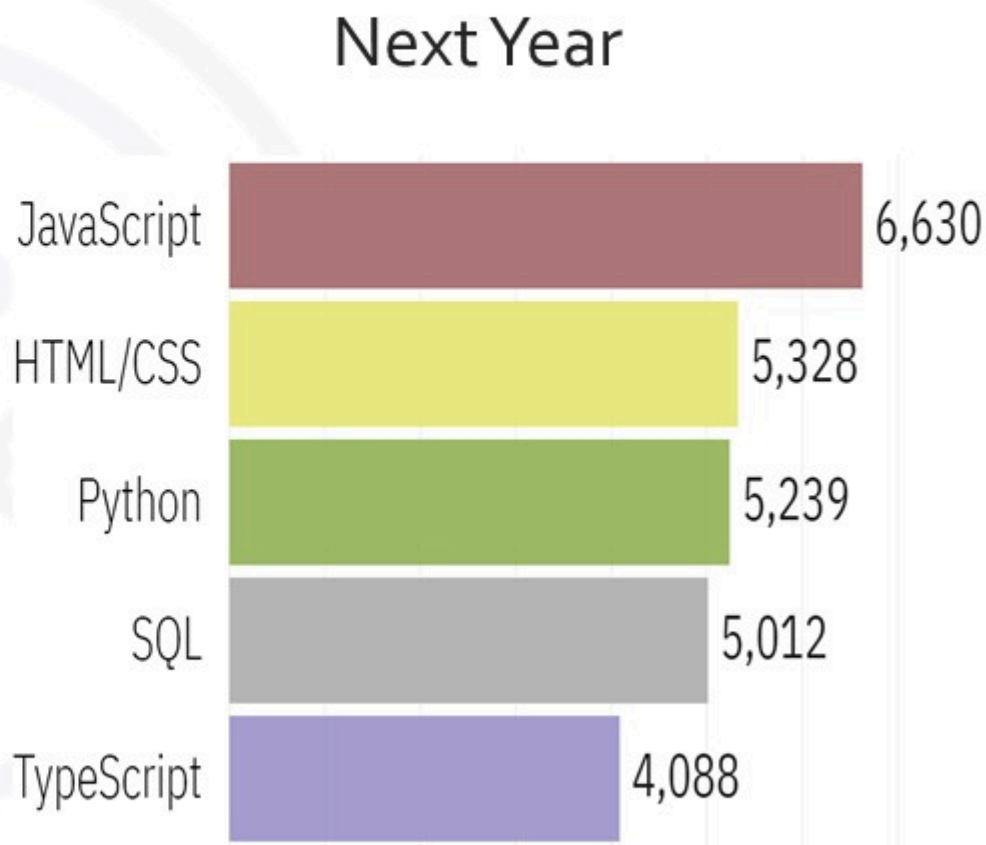
- Data Sources:
  - Stack Overflow Survey 2019
  - Github Job Posting
  - Programming Language Annual Salary
- Data Exploration and Cleaning
  - Python
- Data Visualization
  - IBM CognosAnalytics
- Presentation

RESULTS

- Dataset before cleaning rows - 11552, columns – 85, duplicate rows – 154, after removing N= 11398.
- Median age of the respondents – 29 years and 75 % of them are under 35. Min and max ages are 16 and 99 years. The age group of 24-32 years has the highest number of survey respondents.
- Majority of respondents are male 93.5 % vs 6.5% - female.
- Educational level – Bachelor's degree - 51.8%, Master's degree-23.9%.
- Of professional developers who studied at the university level, over 60% engineering, or software engineering.
- 96 % are employed full-time and 4% - are part-time.
- 90 % of the respondents are developers by profession and 10% are not primarily developers, but write code from time to time database administrators, data or business analysts, data scientists or machine learning specialists, designers, system administrators, engineers (, researchers).
- Median salary in the industry after removing outliers: $52,704 per year. mean compensation – $34,567.
- Distribution of salaries depending on the coding language (last slide): maximum - Swift (for IOS development), then Python, C++, JavaScript.
- Job postings demonstrate the demand for specialists, regarding their skills - most wanted are with a qualification in
- Converted compensation has the biggest correlation with age – which means older (often more qualified, with higher job positions) developers have higher compensations.

PROGRAMMING LANGUAGE TRENDS





PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

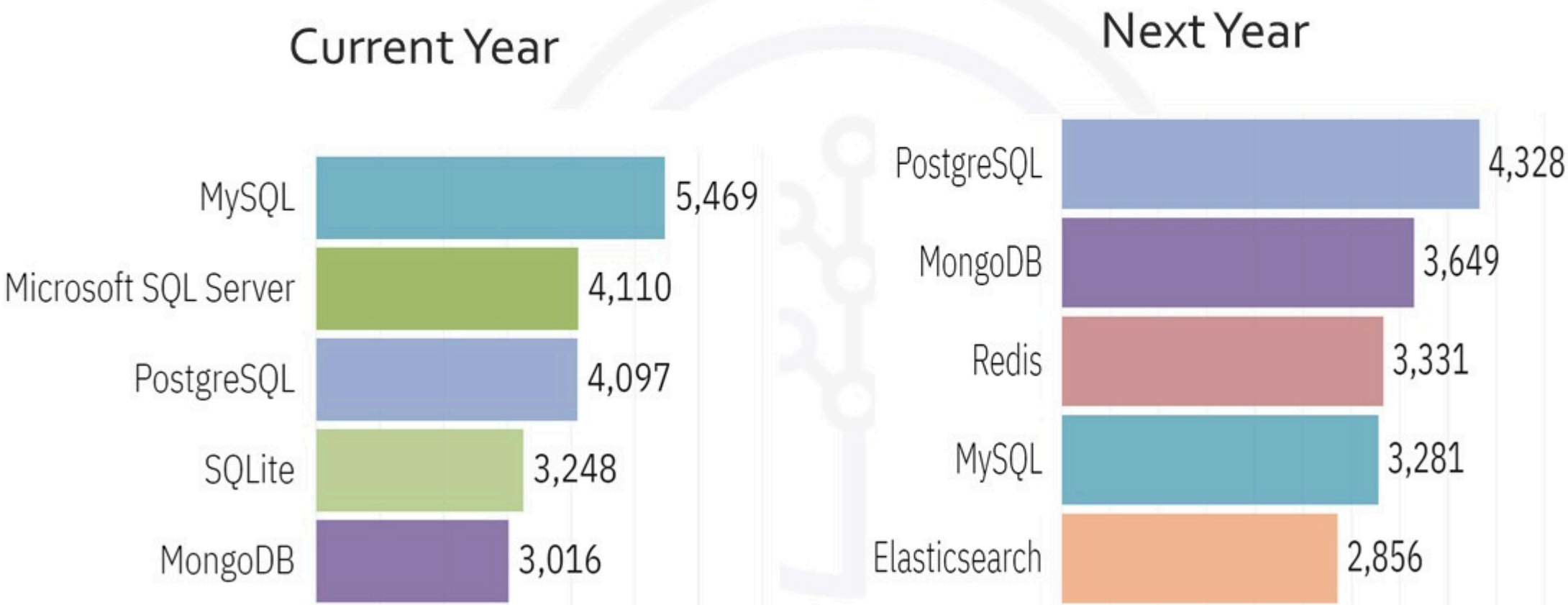
Findings

- JavaScript - The most demanded programming language.
- Other- Python, HTML/CSS, and SQL
- TypeScript more desirable than Bash/Shell/PowerShell.

Implications

- Python's increase may surpass JavaScript in the future.
- Consistent demand for HTML/CSS, Java, SQL, and Python.
- High demand for Web Development

DATABASE TRENDS



DATABASE TRENDS - FINDINGS & IMPLICATIONS

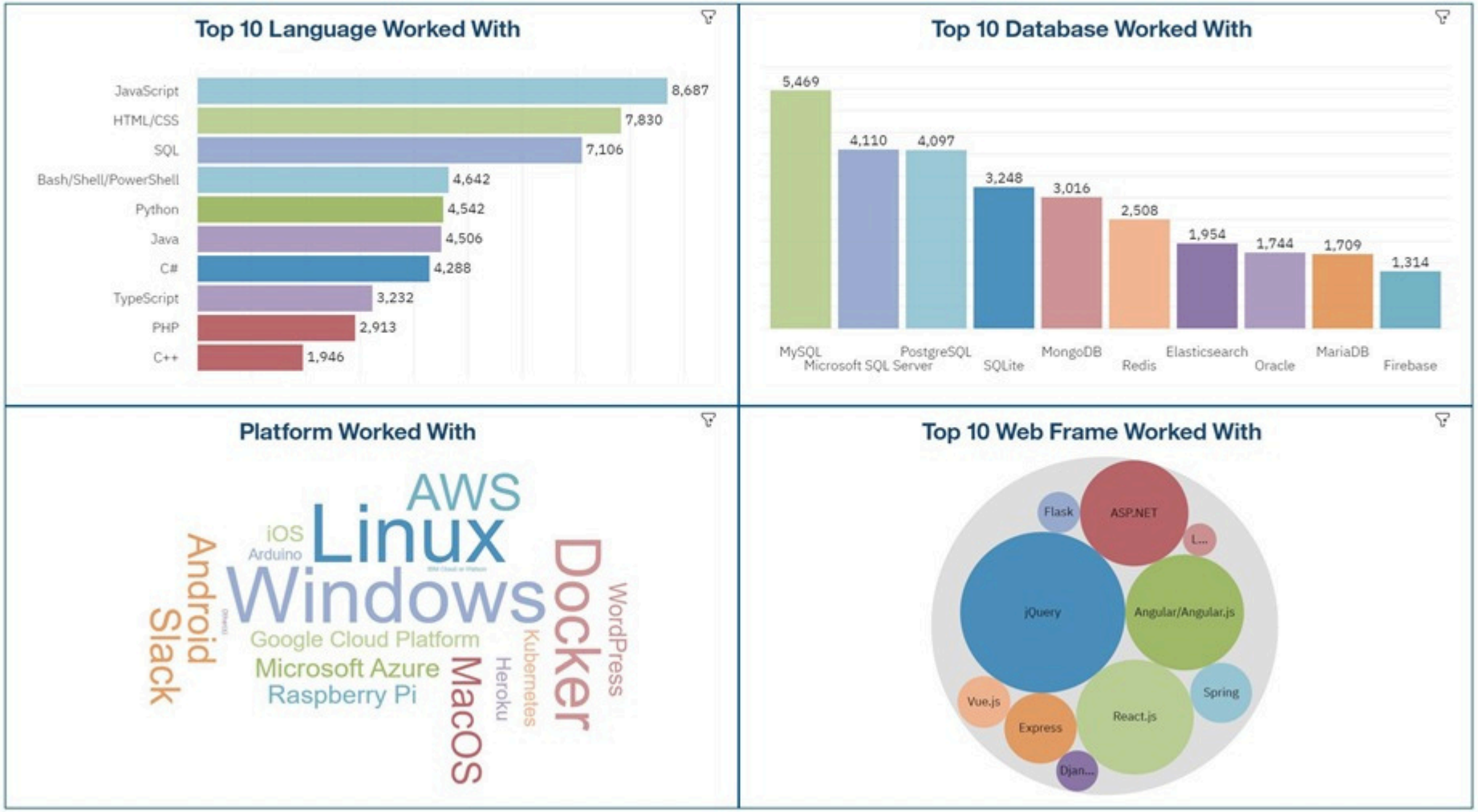
Findings

- Demand increase: PostgreSQL - MongoDB
- Demand decrease: MySQL
- Desire increase: Redis - ElasticSearch

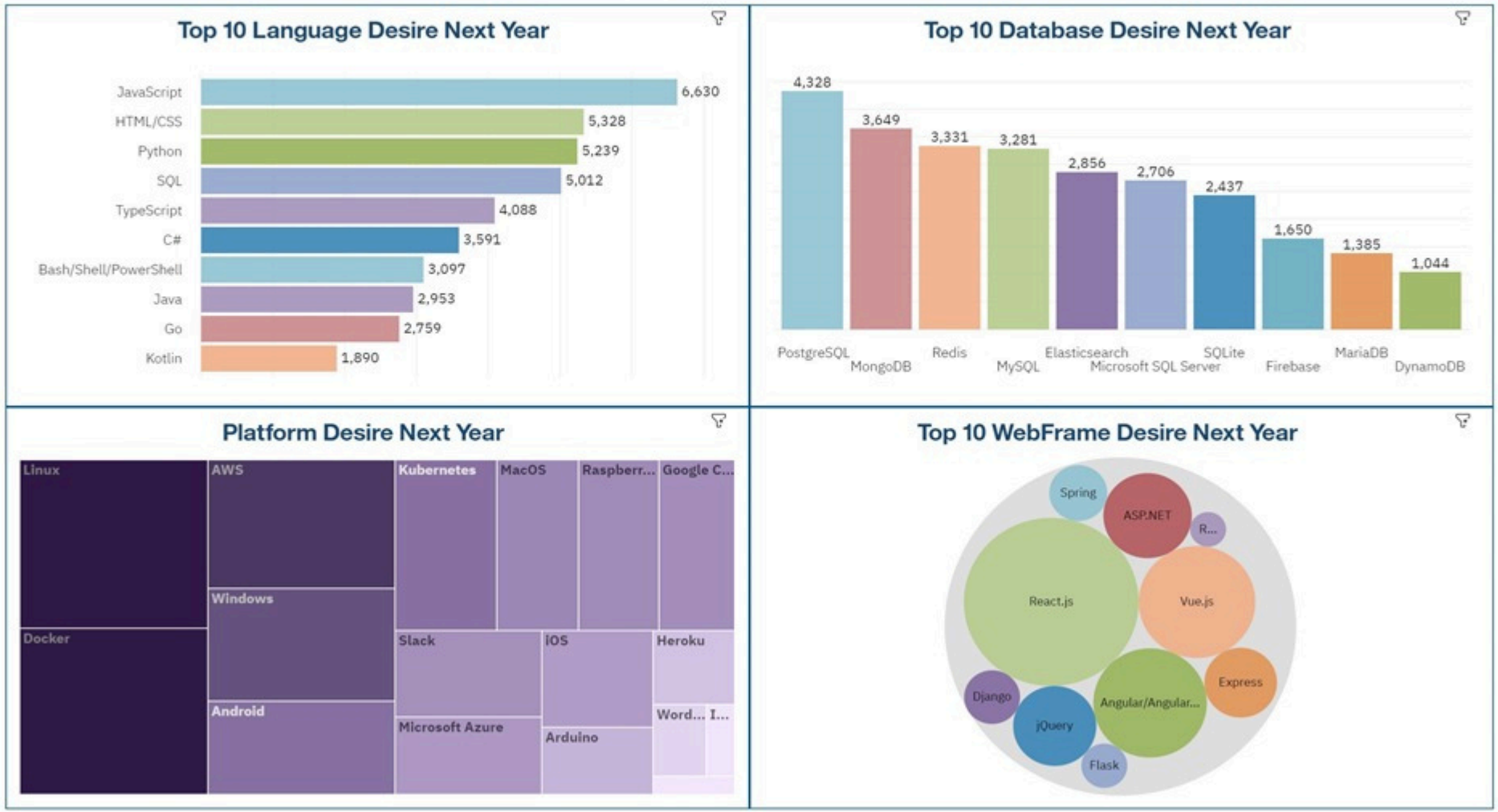
Implications

- Demand consistency: PostgreSQL - MongoDB
- Preferred databases: Open Source

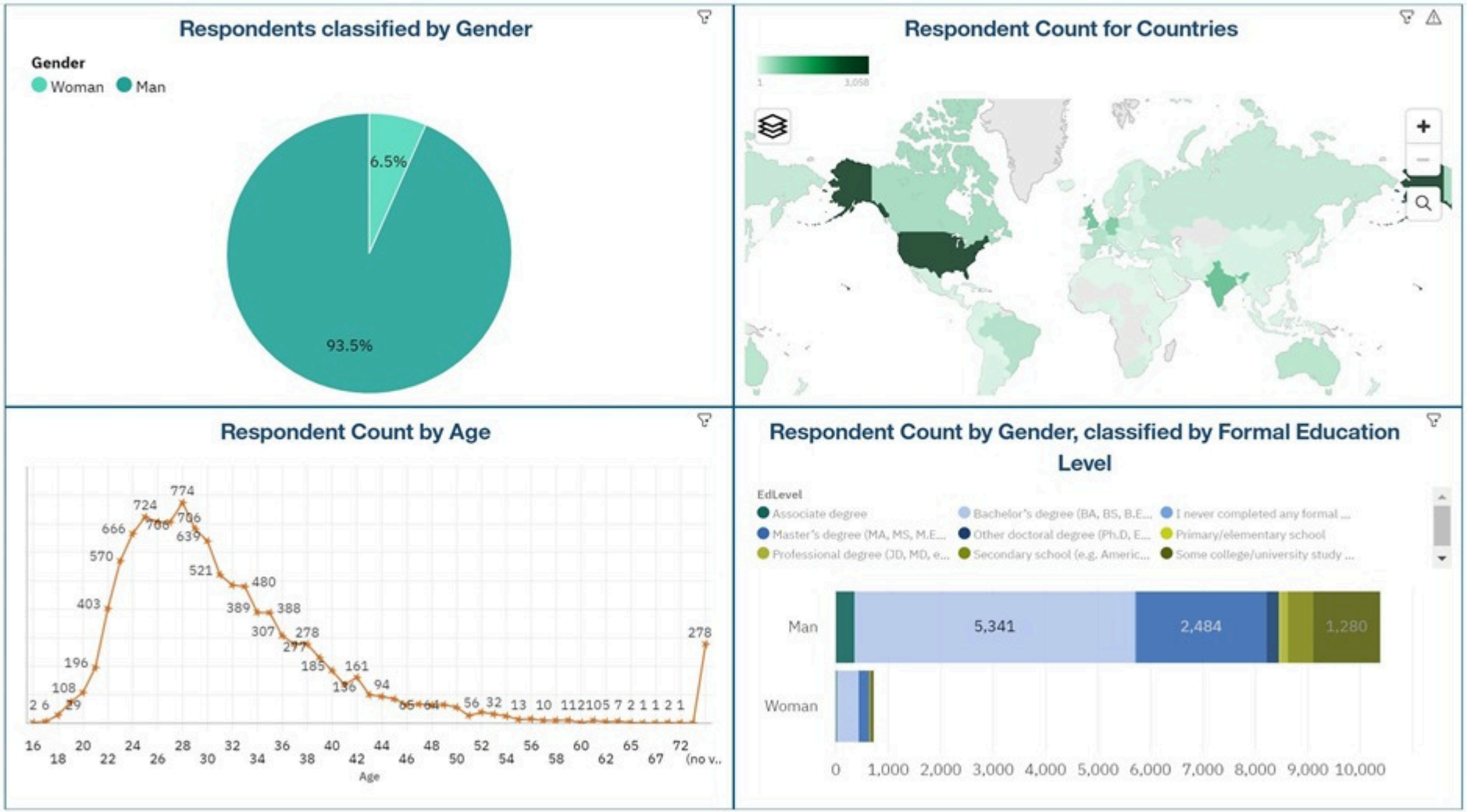
DASHBOARD TAB.1 Current Technology Usage



DASHBOARD TAB.2 Future Technology Trend



DASHBOARD TAB.3 Demographics



DISCUSSION



The insights and trends include the following questions:

- *What are the top programming languages in demand?*
- *What are the top databases in demand?*
- *What are the most popular developers technologies, web frames, platforms?*

FINDINGS & IMPLICATIONS

Findings

- Demands and trends can vary based on new technologies.
- Popular Languages: JavaScript - Python
- Popular Databases: PostgreSQL

Implications

- Companies and individuals need to adapt the new demands.
- Proficiency in higher demand skills can lead to more opportunities.

CONCLUSION



- New technologies lead to varied trends and demands.
- Most popular programming Languages: JavaScript - Python
- Most popular database: PostgreSQL
- The Information Technology field needs diversity.